| INCH-POUND |

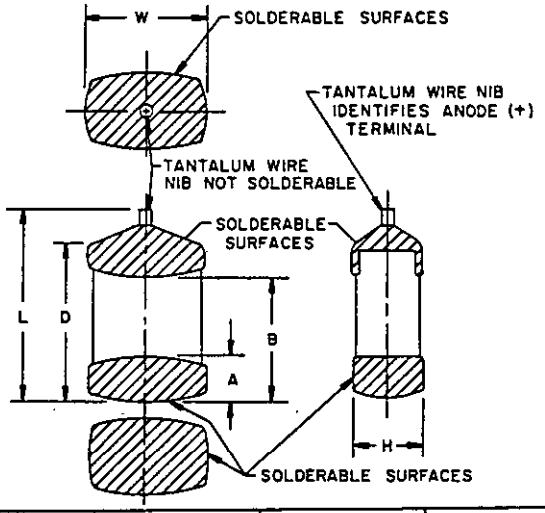
MIL-C-55365/7A
23 August 1990
SUPERSEDING
MIL-C-55365/7
10 March 1987

MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, ESTABLISHED RELIABILITY, STYLE CWR10

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55365.



Inches	mm	Inches	mm
0.005	0.13	0.069	1.75
0.008	0.20	0.072	1.83
0.010	0.25	0.079	2.01
0.015	0.38	0.085	2.16
0.018	0.46	0.104	2.64
0.020	0.51	0.115	2.92
0.027	0.69	0.136	3.45
0.040	1.02	0.138	3.51
0.045	1.14	0.200	5.08
0.048	1.22	0.235	5.97
0.050	1.27	0.243	6.17
0.051	1.30	0.273	6.93
0.063	1.60	0.275	6.98

Size code	L	W	H	A	B	D
C	.079 ±.008	.045 ±.010	.045 ±.010	.018 ±.008	.050 ±.010	.063 ±.008
R	.273 ±.008	.235 ±.010	.136 ±.010	.040 ±.020	.200 ±.027	.243 ±.010
S	.138 ±.005	.072 ±.008	.048 ±.008	.020 ±.010	.085 ±.015	.115 ±.010
V	.138 ±.005	.104 ±.010	.051 ±.010	.020 ±.010	.085 ±.015	.115 ±.010
X	.275 ±.010	.104 ±.010	.051 ±.010	.040 ±.020	.200 ±.027	.243 ±.010
Y	.275 ±.010	.104 ±.010	.069 ±.010	.040 ±.020	.200 ±.027	.243 ±.010
Z	.275 ±.010	.104 ±.010	.104 ±.010	.040 ±.020	.200 ±.027	.243 ±.010

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These capacitors are designed for mounting by dip soldering, vapor phase reflow soldering, of other conventional means.
4. These capacitors will have a minimum of two bonding pads (top and bottom), on the major (width) dimension.
5. The anode termination (dimension D minus dimension B) shall be .010 inch (0.25 mm) minimum.

FIGURE 1. Style CWR10 (conformally coated) capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Chip.

Material: Sintered tantalum.

Leads: Unleaded.

Termination finish: In accordance with MIL-C-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-C-55365).

Operating temperature range: -55°C to +125°C.

Failure rate level: In accordance with MIL-C-55365.

DC leakage (DCL): See table I.

Capacitance (Cap.): See table I.

Cap. tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Equivalent series resistance (ESR) at 100 kHz: In accordance with MIL-C-55365. See table I.

Resistance to soldering heat: In accordance with MIL-C-55365.

Stability at low and high temperatures: In accordance with MIL-C-55365.

Step 1 (+25°C):	DCL	-	See table I
	Cap.	-	Shall not exceed tolerance of table I value.
	DF	-	See table I.
	ESR	-	See table I.
Step 2 (-55°C):	ΔCap.	-	Shall not exceed ±10 percent of initial measurement.
	DF	-	See table I.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I.
Step 4 (+85°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±10 percent of initial +25°C measurement.
	DF	-	See table I.
Step 5 (+125°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±15 percent of initial +25°C measurement.
	DF	-	See table I.

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Step 6 (+25°C): DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial +25°C measurement.
DF - See table I.
ESR - See table I.

Life: Method 108 of MIL-STD-202.

2,000 hours:

At 25°C: DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial measured value.
DF - See table I.
ESR - See table I.

At 85°C: DCL - See table I.

At 125°C: DCL - See table I.

Surge voltage: In accordance with MIL-C-55365.

Part or Identifying Number (PIN): See table I. The term Part or Identifying Number (PIN) is equivalent to the term "type designation" previously used in this specification.

TABLE I. Style CWR10 characteristics.

PIN 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
	volts	µF	µA	µA	µA	%	%	%	ohms	
CWR10C-335--	4	3.3	0.5	5.0	6.0	6	8	8	6.000	C
CWR10C-685--	4	6.8	1.0	10.0	12.0	6	8	8	2.000	S
CWR10C-106--	4	10	1.0	10.0	12.0	6	9	12	2.000	S
CWR10C-156--	4	15	1.0	10.0	12.0	6	9	12	1.500	V
CWR10C-226--	4	22	1.5	15.0	18.0	6	9	12	1.000	X
CWR10C-336--	4	33	3.0	30.0	36.0	6	9	12	1.000	X
CWR10C-476--	4	47	3.0	30.0	36.0	6	9	12	0.600	Y
CWR10C-686--	4	68	3.0	30.0	36.0	6	9	12	0.600	Y
CWR10C-107--	4	100	4.0	40.0	48.0	6	9	12	0.400	Z
CWR10C-337--	4	330	13.5	135.0	162.0	10	12	15	0.080	R
CWR10D-225--	6	2.2	0.5	5.0	6.0	6	8	8	7.000	C
CWR10D-475--	6	4.7	1.0	10.0	12.0	6	8	8	2.000	S
CWR10D-685--	6	6.8	1.0	10.0	12.0	6	8	8	2.000	S
CWR10D-106--	6	10	1.0	10.0	12.0	6	8	10	1.500	V
CWR10D-156--	6	15	1.5	15.0	18.0	6	8	10	1.000	X
CWR10D-226--	6	22	3.0	30.0	36.0	6	9	12	1.000	X
CWR10D-476--	6	47	3.0	30.0	36.0	6	9	12	0.600	Y
CWR10D-107--	6	100	6.0	60.0	72.0	6	9	12	0.400	Z
CWR10D-187--	6	180	13.5	135.0	162.0	10	12	15	0.080	R
CWR10D-227--	6	220	13.5	135.0	162.0	10	12	15	0.080	R

See footnote at end of table.

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TABLE I. Style CWR10 characteristics - Continued.

PIN 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Max ESR 100 kHz	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	+25°C	
	volts	µF	µA	µA	µA	%	%	%	ohms	
CWR10F-155--	10	1.5	0.5	5.0	6.0	6	8	8	10.000	C
CWR10F-335--	10	3.3	1.0	10.0	12.0	6	8	8	2.500	S
CWR10F-475--	10	4.7	1.0	10.0	12.0	6	8	8	2.000	S
CWR10F-685--	10	6.8	1.0	10.0	12.0	6	8	8	2.000	V
CWR10F-106--	10	10	1.5	15.0	18.0	6	8	8	1.500	X
CWR10F-156--	10	15	3.0	30.0	36.0	6	8	8	1.000	X
CWR10F-226--	10	22	3.0	30.0	36.0	6	8	10	0.600	Y
CWR10F-336--	10	33	3.0	30.0	36.0	6	8	10	0.400	Z
CWR10F-476--	10	47	5.0	50.0	60.0	6	8	10	0.400	Z
CWR10F-107--	10	100	10.0	100.0	120.0	10	12	15	0.075	R
CWR10F-127--	10	120	12.0	120.0	144.0	10	12	15	0.070	R
CWR10F-157--	10	150	15.0	150.0	180.0	10	12	15	0.065	R
CWR10H-105--	15	1.0	0.5	5.0	6.0	6	8	8	10.000	C
CWR10H-225--	15	2.2	1.0	10.0	12.0	6	8	8	3.500	S
CWR10H-335--	15	3.3	1.0	10.0	12.0	6	8	8	2.500	S
CWR10H-475--	15	4.7	1.0	10.0	12.0	6	8	8	2.000	V
CWR10H-685--	15	6.8	1.5	15.0	18.0	6	8	8	1.500	X
CWR10H-106--	15	10	1.5	15.0	18.0	6	8	8	1.500	X
CWR10H-156--	15	15	2.0	20.0	24.0	6	8	8	0.600	Y
CWR10H-226--	15	22	3.0	30.0	36.0	6	8	8	0.400	Z
CWR10H-336--	15	33	5.0	50.0	60.0	6	8	8	0.400	Z
CWR10H-686--	15	68	10.5	105.0	126.0	6	8	10	0.095	R
CWR10J-684--	20	.68	0.5	5.0	6.0	6	8	8	12.000	C
CWR10J-155--	20	1.5	1.0	10.0	12.0	6	8	8	3.500	S
CWR10J-225--	20	2.2	1.0	10.0	12.0	4	6	8	3.500	S
CWR10J-335--	20	3.3	1.5	15.0	18.0	4	6	8	3.000	V
CWR10J-475--	20	4.7	1.5	15.0	18.0	4	6	8	1.500	X
CWR10J-685--	20	6.8	2.0	20.0	24.0	6	8	8	1.500	X
CWR10J-106--	20	10	3.0	30.0	36.0	6	8	8	1.000	Y
CWR10J-156--	20	15	3.0	30.0	36.0	6	8	8	0.600	Z
CWR10J-226--	20	22	4.5	45.0	54.0	6	8	8	0.400	Z
CWR10J-476--	20	47	9.5	95.0	114.0	6	8	8	0.110	R
CWR10K-474--	25	.47	0.5	5.0	6.0	4	6	8	15.000	C
CWR10K-684--	25	.68	0.5	5.0	6.0	4	6	8	5.000	S
CWR10K-105--	25	1.0	1.0	10.0	12.0	4	6	8	3.500	S
CWR10K-155--	25	1.5	1.0	10.0	12.0	6	8	8	3.500	S
CWR10K-225--	25	2.2	1.0	10.0	12.0	4	6	8	2.000	V
CWR10K-335--	25	3.3	1.0	10.0	12.0	6	8	8	1.500	X
CWR10K-475--	25	4.7	12.0	120.0	144.0	6	8	8	1.500	X
CWR10K-685--	25	6.8	3.0	30.0	36.0	6	8	8	1.000	Y
CWR10K-106--	25	10	2.5	25.0	30.0	6	8	8	1.000	Y
CWR10K-156--	25	15	4.0	40.0	48.0	6	8	8	0.600	Z
CWR10K-336--	25	33	8.5	85.0	102.0	6	8	8	0.130	R
CWR10M-104--	35	.1	0.5	5.0	6.0	4	6	8	20.000	S

See footnote at end of table.

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TABLE I. Style CWR10 characteristics - Continued.

PIN 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Max ESR 100 kHz	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	+25°C	
	volts	µF	µA	µA	µA	%	%	%	ohms	
CWR10M-154--	35	.15	0.5	5.0	6.0	4	6	8	18.000	S
CWR10M-224--	35	.22	0.5	5.0	6.0	4	6	8	15.000	S
CWR10M-334--	35	.33	0.5	5.0	6.0	4	6	8	12.000	S
CWR10M-474--	35	.47	1.0	10.0	12.0	4	6	8	9.000	S
CWR10M-684--	35	.68	1.0	10.0	12.0	4	6	8	5.000	S
CWR10M-105--	35	1.0	1.0	10.0	12.0	4	6	8	3.500	S
CWR10M-155--	35	1.5	1.0	10.0	12.0	4	6	8	3.000	V
CWR10M-225--	35	2.2	2.0	20.0	24.0	6	8	8	2.000	X
CWR10M-335--	35	3.3	2.0	20.0	24.0	4	6	8	1.500	Y
CWR10M-475--	35	4.7	3.0	30.0	36.0	6	8	8	0.800	Z
CWR10M-685--	35	6.8	2.5	25.0	30.0	6	8	8	0.800	Z
CWR10M-106--	35	10	3.5	35.0	42.0	6	8	8	0.600	Z
CWR10M-156--	35	15	5.5	55.0	66.0	6	8	8	0.190	R
CWR10M-226--	35	22	8.0	80.0	96.0	6	8	8	0.160	R

1/ Complete PIN shall include additional symbols to indicate termination finish, capacitance tolerance, and failure rate level.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 17, 99
NASA - NA
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 11, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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